

STATEMENT OF WORK

1. Objective/Requirements

NASA Langley Research Center requires the purchase of Speedgoat IO Modules which are needed for interfacing with avionics components as part of the SWIFT GMATT flight systems.

2. Characteristics, Scope, and Specs

Part #	Description	Qty
2A1455	IO145-Mobile Fast-sampling 16-bit analog I/O module with current and high voltage signal conditioning: - 16 x single-ended or 8 x differential sequential sampling analog inputs supporting voltage ranges up to ± 48 V (SE) or ± 96 V (DF). - 8 x analog inputs supporting a current range of up to ± 25 mA. - 8 x analog outputs supporting current ranges up to 24 mA and voltage ranges of up to ± 12 V, individually configurable by channel. - 8 x 3.3 V (5 V-tolerant) digital TTL I/O lines individually configurable by channel. Includes Speedgoat driver blockset for R2024a, I/O cable, and terminal board. Installed in Mobile real-time target machine and fully tested.	1
2A3065	IO306-Mobile Configurable I/O module with 64 x 3.3V TTL I/O lines and Xilinx Spartan 6 FPGA with 24k logic cells. Includes Speedgoat driver blockset for R2024a, I/O cable, and terminal board. Installed in Mobile real-time target machine and fully tested.	1
20306Z	IO306-Configuration Package The I/O configuration package includes 3 configurations: 1.) Configuration for rapid control prototyping: 16 PWM channels, 3 CAP channels, 4 QAD channels, 4 Pulse Counter channels and 1 interrupt line. All I/Os can also be used as digital I/O 2.) Configuration for hardware in-the-loop: 3 PWM channels, 16 CAP channels, 4 QAE channels, 4 Pulse Counter channels and 1 interrupt line. All I/Os can also be used as digital I/O 3.) Configuration for communication: 4 SPI interfaces, 4 I2C master interfaces, 4 I2C slave interfaces, 6 serial (UART) interfaces and 1 interrupt line. All I/Os can also be used as digital I/O	1
205045	IO504-Mobile 8-port RS232/RS422/RS485 I/O module for baud rates up to 920 kb/s (RS232) and 5.5296 Mb/s (RS422/485). Includes Speedgoat driver blockset for R2024a, I/O cable, and terminal board. Installed in Mobile real-time target machine and fully tested.	1
2B6035	IO603-4-T-Mobile Intelligent CAN I/O module with four independent galvanic isolated ports. Each port is software configurable to either support CAN FD or CAN HS. In addition, the IO603-4 provides 2 x LIN Master/Slave channels and 4 digital input and output lines. Access to all I/O is via a 25-pin DSUB connector or via 5x 9-pin DSUB (male) connectors also included in the delivery, as well as CAN loopback test cables. With software-configurable 120-ohm CAN bus terminations. Extended temperature range: -40° C to $+75^{\circ}$ C	1

	Includes Speedgoat driver blockset for R2024a. Installed in Mobile real-time target machine and fully tested.	
401099	Shipping & Handling	1
307500	Level One Hardware Warranty Service The Level One Hardware Warranty Service includes 36 months (3 years) of warranty on all hardware components. During warranty period, defective hardware is repaired at no charge. Speedgoat offers timely repair services by trained factory technicians.	1

3. Place of Performance

NASA Langley Research Center
4 Langley Blvd.

[REDACTED]

[REDACTED]

Hampton, VA 23681

[REDACTED]

4. Period of Performance

4 weeks ARO